Walt Disney and Computer User Interface Design

Harry R. Chesley

Tuesday, December 13, 1988

"Animation can explain whatever the mind can conceive."
Walt Disney

Prologue

In ten years, maybe even five, we'll look back on the Macintosh user interface and wonder how people put up with something so static, wooden, non-intuitive, and ugly. I think we all accept that fact at Apple, but the fact alone doesn't provide us with much insight into how the next interface will look and operate. It does, however, provide us with the motivation to look for something newer and better, and the expectation that we'll find something if we look hard enough.

This paper is about the lessons we can learn from another group of people who faced pretty much the same problem roughly sixty years ago: Walt Disney Productions. When Walt Disney began, in 1923, animation was a fairly new technology. It was rough, wooden, and simplistic (with one or two notable exceptions). When Disney and his animators finished with it, animation was an art form. Besides the dynamics of a group of people trying to invent a new form of communication (which I won't dwell on here), there are much more specific lessons we can gather from their experience: Both animation and computers are a communications medium, both use graphics as the primary communications tool, augmented with sound, both need to ensnare, entertain, and enlighten the viewer. In fact, as we move into the next generation of computer hardware, we will have the power to routinely generate and display animation as a part of the computer user interface. But computers also allow interaction with the user, which animation does not.

In the long run — the hundred year perspective — I expect film animation will be seen as a short but significant prelude to computer systems, in terms of communications tool, entertainment device, and business aide. It is therefore obvious and natural that we build directly on the work done by the animators that came before us, even though their medium was slightly different than ours.

The Book

Of the existing animators and animation studios, Disney is clearly the leader. We are extremely fortunate that the Disney studios has published a book chronicling their history and describing in full detail the central principles of their work. This book is:

Disney Animation — The Illusion of Life
by Frank Thomas & Ollie Johnston

Anyone with any serious interest in the next generation of computer user interfaces should read this book.

The Book interweaves a history of Walt Disney Productions together with the techniques used to produce the animation. It describes the genesis of all of the well known Disney films, and contains 489 really tremendous full color plates. It tells the history of the Disney studios. It explains the techniques used, both the abstract, conceptual approaches (which are the most useful to us at Apple), and the day-to-day logistics of getting a full-length animation produced.

The Principles

Chapter 3 of The Book lists the twelve basic principles of Disney animation. This are something like our Macintosh User Interface Guidelines, but more abstract and distilled. The twelve principles are:

1. Squash & stretch.
2. Anticipation
3. Staging
4. Straight ahead action and pose to pose
5. Follow through and overlapping action
6. Slow in and slow out
7. Arcs
8. Secondary action
9. Timing
10. Exaggeration
11. Solid drawing
12. Appeal

In order to show how relevant these principles are to our work, I will explain each and give examples of how a "Disney Finder" might make use of the principle. (A "Disney Finder" per se might even make sense as a joint Apple/Disney product for the next generation K-12 machine.)

Many of these principles are related to making the animation more "life like." Before the reader has a chance to say "but I don't necessarily want my computer to be alive," let me say that these principles also apply to making the system more dynamic, enthralling, exciting, and compelling; more "animated."

1. Squash & Stretch

Few things in the "real world" move rigidly. Some part of an object lags behind when movement is begun, and continues on after it is halted. In Disney animation, this effect is shown by stretching objects or characters as they start to move, and by having them squash as they stop or run into something.

In the Finder, squash and stretch could be used any time an object is moved. When you "grab" an icon with the mouse and move it around the screen, it would stretch out as the movement begins, and squash to a halt when you stop moving the icon. This effect would yield a much more fluid, dynamic feeling to the objects in the system, and to the system as a whole.

2. Anticipation

If an action occurs on the screen without any warning, the audience will often miss it, or perceive only a part of it. Anticipation is the principle of preceding the main action with another action that leads the audience to expect the main action. For example, having a character look at an object before picking it up.

In the Finder, we use anticipation today in the zooming that presages the appearance of a folder window (this is also partly staging, as described next). We do not, however, use anticipation at all in the case of a window that opens in response to selecting a desk accessory. It simply appears without warning. In a Disney Finder, you might expect Tinker Bell to fly out from the Apple logo and flick her wand at the point where the window appears.

3. Staging

This is one of the broadest and oldest of the principles, going back essentially to the earliest beginnings of theater. It is the principle that each idea presented must be presented so that it is completely and unmistakably clear. This is done in a wide variety of ways with a wide range of techniques. The choice of perspective angle and width is used extensively in animation and film as a tool for staging.

Today in the Finder, when a series of files is being copied, we use a text display to inform the user of the current activities. A better staged display might have each file "fly" out of the source disk or folder up to the screen, then back to the destination.

Today we don't use changes in perspective at all. This technique might be used in future hardware systems to zoom in on the result of a Find File search, or to provide transitions between normal and fatbits views of bitmaps.

4. Straight Ahead Action and Pose to Pose

These terms apply to approaches to producing an animated sequence. In straight ahead action, the animator simply starts at the beginning and generates drawings, with the direction of the action frequently changing in mid-course in unanticipated directions. In pose to pose, on the other hand, key frames of the sequence are drawn first, then the "in betweens" are drawn in. Straight ahead action drawing tends to be more creative, while pose to pose tends to be more precise.

At first glance, these seem to be animation techniques which offer little of interest to the computer interface designer. However, this is not true at all. The straight ahead approach corresponds to a user-directed sequence, while the pose to pose approach corresponds to a pre-scripted sequence.

When the user writes a script for some action (using Macro Maker, for example), the user is specifying the sequence of poses to be used. The system can then connect them to produce a full, smooth-flowing animation. For example, if I create a macro that selects a file, then chooses a command from the menu, the system could produce an on-screen character — one of the seven dwarves, perhaps — that walks over to the file, touches it, then walks up to the menu, pulls it down and selects the command. This potentially complicated sequence of animation was chosen by the user simply by selecting the actions (poses) to be done in series.

5. Follow Through and Overlapping Action

In life, few things happen in isolation. One action does not come to a complete stop before the next one begins. This principle states that the same should be true in animation. Actions should follow through (like a golf swing), and they should overlap with the next action in a sequence.

Returning to the example of a block of files being transferred, each one "flying" out of the source and into the destination, overlapping action clearly applies here. Although each file should pause in flight to let the user see its name, the next file should already be "in the air" while the current one is being shown, and before the previous one disappears.

And since the computer, unlike animation, is an interactive environment, an even more powerful application of overlapping action is the ability of the user to start up a new action before the previous action is finished.

6. Slow In and Slow Out

This is the technique of moving quickly between key images, and then slowing down in the vicinity of the images themselves. It gives animations a "zippy" feeling, so

long as it isn't overused.

Besides the feeling that this technique provides, it has two other important effects: First, it emphasizes the most important sequences of the animation, drawing the viewers attention to them and giving the viewer time to absorb them. Second, it avoids wasting the viewer's time with long, largely irrelevant intermediate action sequences. The in between sequences lead the viewer, but provide no new information; they supply context and continuity rather than content.

In the Finder, you might imagine a file icon zipping from the folder to hover over the trash can and then diving in. This makes it very clear to the user what action is taking place.

7. Arcs

In the "real world," action never takes place in a straight line. It always moves in an arc, at least when its performed by a living creature. This principle simply states that animated movement should also take place in arcs.

In the computer, we already have a lot of arcing because the user is manipulating the cursor (any time you leave your elbow fixed and swing the cursor across the screen, you're moving it in an arc, whether you notice it or not). But this principle could be applied in many other areas — zooming from icon to windows for example. And it will have many more application when agents are personified on the screen.

8. Secondary Action

Secondary action is the use of subsidiary movement or ideas to reinforce the principle action. In traditional animation, for example, it might, be a character wiping away a tear as he turns dejectedly away.

Taking the example of a file descending into the trash can, a secondary action might be the trash can opening its lid in anticipation, or the file icon "graying out" as it descends.

9. Timing

This is the issue of how much time is spent in each part of a sequence of action. It is extremely important to the flow of action in a sequence.

In computers, timing becomes even more complex. Besides needing to consider the same issues that traditional animators consider, design of computer animation has three additional complications: First, a given sequence will be seen over and over. What was interesting and informative the first time, may be excruciatingly slow and boring the 100th time. Second, the computer is interactive, and actions on the part of the user need to be factored into the equation of timing. Third, the computer is constrained to the timing of the hardware; it cannot make a file copy take less time that it does, though it can make it take longer.

10. Exaggeration

This is the technique of making animation more "real" than reality, of making it a caricature of reality. This makes the animation more alive than it would be otherwise.

Caricature could be used in the Macintosh note, warning, and error dialog boxes simply by picking the right Disney character to replace the current icons. Depending upon the severity and type of error (user versus system), Scrooge MacDuck or Goofy might appear, whereas a note could be delivered by a bright, helpful Mickey Mouse.

11. Solid Drawing

Solid drawing is simply good artwork. This is a requirement for any task that involves art, but is especially crucial in animation, where so many drawings are used.

The computer can help generate the art, can act as librarian, and can help produce the variations that are needed from sequence to sequence and within sequences. But it cannot replace the artist. This is obvious today if you look at HyperCard stacks. It is clear which ones were done by professional artists, and which ones where done a amateurs.

12. Appeal

This principle states that the animation should be attractive, appealing, pleasant to the viewer. In Disney animation, even the evil characters are visually appealing in one way or another.

This principle should be obvious to anyone who works on or with the Macintosh. It was this, in large part, that made the Macintosh such a success in the early days. Speaking for myself, I know that my first, gut level reaction to the Mac user interface was "I *like* this," followed shortly by "I want one of these."

The Future

I have tried to stay within the context of the existing Macintosh user interface in creating the examples given above. I did this intentionally in order to provide a clear, simple bridge for people familiar with the current Mac. However, the real power of these concepts will not come with the existing interface, it will come with the next generation interface (indeed, I had to stretch quite a bit sometimes to find examples within the current context, because it was designed specifically without animation as a tool).

The next generation system will have the following characteristics:

- Easy access to much more complicated information structures.
- A visual context with more depth (cyberspace), where the current desktop is a special case.
- Wide bandwidth connectivity to other systems and users.
- Multiple, user-driven actions occurring at the same time.
- Agents which are not directly user driven from moment to moment.
- User programming.

In order to convey the magnitude and diversity of information implied by these elements, animation is more than just a pretty convenience, it's a necessity. And, of course, animation will also be used for content as well as interface. In this world, the techniques enumerated in the paper, and discussed in much greater detail in the *Disney Animation* book, will be the fundamental principles of the next generation system.

Post Script

You will have noticed that this paper includes no graphics, and may think this strange. There are two reasons: (1) I wanted to get this out quickly, and graphics take time, especially for a non-artist such as myself. (2) The examples to be found in the *Disney Animation* book are so tremendous that mine would inevitably pale besides them. So go buy the book!

Note to Cupertino readers: as of 12/13/88, A Clean, Well Lighted Place for Books has a whole stack of them on sale.